

Structure of solutions of linear ODEs, Continued

1. Nonhomogeneous linear ODEs

Consider

$$y^{(n)} + a_{n-1}(t)y^{(n-1)} + \cdots + a_1(t)\dot{y} + a_0(t)y = g(t). \quad (1)$$

Assume that $a_0, \dots, a_{n-1}$ and g are continuous on $I = (a, b)$. What is the consequence of this assumption?

Suppose y_p is a particular solution of ODE (1), and $\{y_1, \dots, y_n\}$ is a fundamental set of solutions (FSS) of the associated homogeneous equation.

Theorem 12.1 (Structure of the general solution). The general solution of ODE (1) has the form

$$y(t) = y_p(t) + \underbrace{\hspace{10em}}_{y_c(t)}, \quad c_i \in \mathbb{R}. \quad (2)$$

where $\{y_1, \dots, y_n\}$ forms a FSS of the Homogeneous version of ODE (1).

The constants c_i are determined by the _____.

Guided proof.

y_p is a solution of ODE (1). So, subbing y_p in (1) must satisfy the equation, i.e.,

_____ (3)

Let y be an arbitrary solution of ODE (1). So, subbing y in (1) must satisfy the equation, i.e.,

_____ (4)

Subtract Equation (3) from (4).

_____ (5)

What does it tell us about function $y_c = (y - y_p)$?

So, for any solution y , $y_c = (y - y_p) = \sum_{i=1}^n c_i y_i$ where $\{y_1, \dots, y_n\}$ forms an FSS of Homogeneous version of the ODE (1).

1.1. How to determine the coefficients c_i ?

For the IVP with initial time $t_0 \in I$ and initial condition $y^{(j)}(t_0) = \eta_j$, $j = 0, \dots, n-1$ with particular solution $y_p(t)$, the solution y_c of the homogeneous equation must satisfy

$$y^{(j)}(t_0) = \eta_j \implies y_c^{(j)}(t_0) + y_p^{(j)}(t_0) = \eta_j \implies y_c^{(j)}(t_0) = \underline{\hspace{2cm}}, \quad j = 0, \dots, n-1.$$

Write the equation above in vector form:

$$\begin{pmatrix} y_c(t_0) \\ \dot{y}_c(t_0) \\ \vdots \\ y_c^{(n-1)}(t_0) \end{pmatrix} = \underline{\hspace{2cm}}, \quad \text{where } \boldsymbol{\eta} = \begin{pmatrix} \eta_0 \\ \eta_1 \\ \vdots \\ \eta_{n-1} \end{pmatrix}, \quad \mathbf{p}_0 = \begin{pmatrix} y_p(t_0) \\ \dot{y}_p(t_0) \\ \vdots \\ y_p^{(n-1)}(t_0) \end{pmatrix}.$$

Write the system that determines the constants.

Define $\mathbf{c} = (c_1, \dots, c_n)^\top$, and

$$M_0 = \begin{pmatrix} y_1(t_0) & \cdots & y_n(t_0) \\ \dot{y}_1(t_0) & \cdots & \dot{y}_n(t_0) \\ \vdots & \ddots & \vdots \\ y_1^{(n-1)}(t_0) & \cdots & y_n^{(n-1)}(t_0) \end{pmatrix}.$$

Write $(y_c(t_0), \dots, y_c^{(n-1)}(t_0))^\top$ in terms of M_0 and $\mathbf{c}$.

$$\begin{pmatrix} y_c(t_0) \\ \dot{y}_c(t_0) \\ \vdots \\ y_c^{(n-1)}(t_0) \end{pmatrix} = \begin{pmatrix} c_1 y_1(t_0) + \cdots + c_n y_n(t_0) \\ c_1 \dot{y}_1(t_0) + \cdots + c_n \dot{y}_n(t_0) \\ \vdots \\ c_1 y_1^{(n-1)}(t_0) + \cdots + c_n y_n^{(n-1)}(t_0) \end{pmatrix} = \begin{pmatrix} \underline{\hspace{2cm}} & \cdots & \underline{\hspace{2cm}} \\ \vdots & \ddots & \vdots \\ \underline{\hspace{2cm}} & \cdots & \underline{\hspace{2cm}} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = \underline{\hspace{2cm}}$$

Now, find $\mathbf{c} = (c_1, \dots, c_n)^\top$ by imposing the initial conditions.

$$M_0 \mathbf{c} = \underline{\hspace{2cm}} \implies \mathbf{c} = \underline{\hspace{2cm}}.$$

Remark. M_0 is invertible because its columns are the initial vectors of an FSS, i.e., Wronskian.

1.2. Procedure for solving a linear ODE

- Find an FSS for the $\underline{\hspace{2cm}}$.
- Find a $\underline{\hspace{2cm}}$ of (1).
- Set $y(t) = y_p(t) + \sum_{i=1}^n c_i y_i(t)$.
- Impose the initial conditions and solve for $\underline{\hspace{2cm}}$.

2. Linear homogeneous constant-coefficient ODEs

We now consider

$$y^{(n)} + a_{n-1}y^{(n-1)} + \cdots + a_1\dot{y} + a_0y = 0, \quad a_0, \dots, a_{n-1} \in \mathbb{R}. \quad (6)$$

For variable coefficients, explicit solutions are generally difficult to find; integrating factors and reduction of order apply in suitable cases. For constant coefficients, an FSS can be found systematically, **Laplace transforms can give particular solutions in many cases**, and **stability can be tested directly**.

2.1. The Zero Equilibrium

Let $x = (y, \dot{y}, \dots, y^{(n-1)})^\top$. Show that $\bar{x} = 0$ is an equilibrium of ODE (6).

Write the state-space form of ODE (6).

$$\dot{x} = \begin{pmatrix} 0 & 1 & 0 & \cdots & \cdots & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \vdots \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{pmatrix} x = F(x).$$

An equilibrium $\bar{x}$ should make $F(\bar{x}) = 0$. Clearly $\bar{x} = 0$ does that.

In the following sections, we study the special cases $n = 1$ and $n = 2$. In the next part we generalize.

2.2. Linear homogeneous constant-coefficient ODEs with Order $n = 1$.

Solve and classify the zero equilibrium.

Write $\dot{y} + a_0 y = 0$ as $\dot{y} = \lambda y$, where $\lambda = -a_0$. An FSS and the general solution are

$$\text{FSS} = \underline{\hspace{2cm}}, \quad y(t) = \underline{\hspace{2cm}}.$$

Sketch $y(t)$ for the three signs of λ , with the same $y(0) > 0$.

Condition	Stability of $\bar{y} = 0$
$\lambda > 0$	
$\lambda = 0$	
$\lambda < 0$	

Justify your answer for $\lambda = 0$.

How come for $\lambda = 0$, $\bar{y} = 0$ is stable but not asymptotically stable? Isn't this ODE scalar?
 Didn't we say that for scalar ODE, any isolated equilibrium that is stable is also asymptotically stable?

Consider any $\epsilon > 0$. We want $\|y(t) - 0\| < \epsilon$ for all time t . Can we always find a δ for any ϵ such that if initial state $y(0) = Y_0$ is in $B_\delta(0)$ (i.e., δ -neighborhood of the zero-equilibrium) then $\|y(t) - 0\| < \epsilon$?

$$\text{For } \lambda = 0, \quad \forall \epsilon > 0, \quad \|y(0) - 0\| < \delta = \underline{\hspace{2cm}} \implies \|y(t) - 0\| < \epsilon, \forall t.$$

2.3. Linear homogeneous constant-coefficient ODEs with Order $n = 2$ and Characteristic Polynomial with Distinct Real Roots.

Consider

$$\ddot{y} + a_1\dot{y} + a_0y = 0. \quad (7)$$

Test an exponential solution $y(t) = e^{\lambda t}$.

Substitute into (7) and use $e^{\lambda t} \neq 0$ to obtain an informative equation about λ .

The **characteristic polynomial** is $p(\lambda) = \underline{\hspace{10em}}$. Its roots give exponential solutions. Define its discriminant by

$$\Delta = \underline{\hspace{10em}}.$$

Case I: $\Delta > 0$ (distinct real roots).

The roots and corresponding solutions are

$$\lambda_{1,2} = \underline{\hspace{10em}}, \quad y_i(t) = e^{\lambda_i t}, \quad i = 1, 2.$$

Are these solutions an FSS? Compute the Wronskian at $t_0 = 0$.

The general solution is

$$y(t) = \underline{\hspace{10em}}.$$

Classify the zero equilibrium in state space.

The state is $x = (y, \dot{y})^T$ and the zero equilibrium is $\bar{x} = (0, 0)^T$.

Root condition	Stability of $\bar{x} = 0$
$\lambda_1 < 0$ and $\lambda_2 < 0$	
At least one root is positive	
One root is negative; the other is zero	

2.4. Linear homogeneous constant-coefficient ODEs with Order $n = 2$ and Characteristic Polynomial with a repeated real root.

Case II: $\Delta = 0$.

The repeated root is $\lambda_1 = \lambda_2 = \lambda$, where $\lambda = \underline{\hspace{2cm}}$. The characteristic polynomial gives one solution $y_1(t) = e^{\lambda t}$.

Check that $y_2(t) = te^{\lambda t}$ is a second solution.

Compute its first two derivatives and substitute into (7).

Verify that $\{e^{\lambda t}, te^{\lambda t}\}$ is an FSS.

The general solution is

$$y(t) = \underline{\hspace{4cm}}.$$

Classify $\bar{x} = (0, 0)^\top$ for each sign of λ .

Condition	Stability of $\bar{x} = 0$
$\lambda < 0$	
$\lambda = 0$	
$\lambda > 0$	

Explain carefully why the repeated root $\lambda = 0$ gives instability.

Complete the state trajectory:

$$x(t) = \begin{pmatrix} y(t) \\ \dot{y}(t) \end{pmatrix} = \underline{\hspace{4cm}}.$$

2.5. Linear homogeneous constant-coefficient ODEs with Order $n = 2$ and Characteristic Polynomial with complex conjugate pair of roots.

Case III: $\Delta < 0$.

Write

$$\lambda_{1,2} = -\frac{a_1}{2} \pm i\frac{\sqrt{-\Delta}}{2} = a \pm ib, \quad a = -\frac{a_1}{2}, \quad b = \frac{\sqrt{-\Delta}}{2} > 0.$$

The two complex solutions are

$$\hat{y}_{1,2}(t) = e^{(a \pm ib)t} = e^{at}(\cos(bt) \pm i \sin(bt)).$$

Both their real and imaginary parts are real solutions. Thus take

$$y_1(t) = \underline{\hspace{2cm}}, \quad y_2(t) = \underline{\hspace{2cm}}.$$

Is this an FSS? Compute the Wronskian at $t_0 = 0$.

The real general solution is

$$y(t) = \underline{\hspace{4cm}}.$$

Classify the zero equilibrium using the real part a .

Condition	Stability of $\bar{x} = (0, 0)^T$
$a = 0$	
$a < 0$	
$a > 0$	

Sketch nearby orbits in the $(y, \dot{y})$ plane. Indicate direction.

